

1711.01340 Problem 5: Literature-Already-Answered Status

FA/Banach discovery run `fa_banach_001`

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Status

This packet records that published Problem 5 in Motakis–Puglisi–Tolias, *Algebras of diagonal operators of the form scalar-plus-compact are Calkin algebras*, arXiv:1711.01340, is already answered affirmatively by Motakis–Pelczar-Barwacz, *Reflexive Calkin algebras*, arXiv:2401.18037.

Original Problem

Problem 5 asks for a Banach space X with an unconditional basis not equivalent to the unit vector basis of c_0 such that the unitization of X is isomorphic, as a Banach algebra, to a Calkin algebra.

morphic, as a Banach algebra to $C(K)$ for an uncountable compact topological space K ?

Remark 8.8. In a personal communication with the first author T. Kania asked whether the unitization of Schreier space from [Schr] endowed with coordinate-wise multiplication with respect to its standard unconditional basis is a Calkin algebra. Our paper does not provide an answer to this. More generally, we observe that our method does not work directly to show that the unitization of an arbitrary Banach space with an unconditional basis is a Calkin algebra. If X is a Banach space with an unconditional basis $(x_i)_i$ and there exists a Banach space Y with a basis so that the unitization of X is isomorphic as a Banach algebra to $\mathbb{R}I \oplus \mathcal{K}_{\text{diag}}(Y)$ then $(x_i)_i$ is equivalent to the unit vector basis of c_0 . Indeed, assume that $T : \mathbb{R}e \oplus X \rightarrow \mathbb{R}I \oplus \mathcal{K}(Y)$ is an algebra isomorphism. Then if $(e_i)_i$ is the basis of Y write for all $k \in \mathbb{N}$ $Tx_k = \text{SOT-}\sum_i a_i^k e_i^* \otimes e_i$. Since $x_k^n = x_k$ for all $n \in \mathbb{N}$ we obtain that in fact there is a subset F_k of $\mathbb{N}$ so that $Tx_k = \text{SOT-}\sum_{i \in F_k} e_i^* \otimes e_i$ and since $x_i x_j = 0$ for $i \neq j$ the sets F_k must be pairwise disjoint. Clearly, $Te = \text{SOT-}\sum_i e_i^* \otimes e_i$. The fact that T is onto implies that each F_k is a singleton $F_k = \{\phi(k)\}$ and that $\cup_k F_k = \mathbb{N}$. If we reorder the basis $(x_k)_k$ as $(x_{\phi^{-1}(k)})_k$ then for all $n \in \mathbb{N}$ $\|\sum_{k=1}^n x_{\phi^{-1}(k)}\| \leq \|T^{-1}\| \|\sum_{k=1}^n e_k^* \otimes e_k\| \leq C\|T^{-1}\|$ where C is the monotone constant of $(e_i)_i$. Unconditionality yields that $(x_i)_i$ is equivalent to the unit vector basis of c_0 .

Problem 5. Find a Banach space X with an unconditional basis $(x_i)_i$ that is not equivalent to the unit vector basis of c_0 so that there exist a Banach space the Calkin algebra of which is isomorphic as a Banach algebra to the unitization of X .

Remark 8.9. The unitization of James space $\mathbb{R}e_\omega \oplus J$ may be viewed as the space of all scalar sequences of bounded quadratic variation and similarly for every Banach space X with a subsymmetric sequence the space $\mathbb{R}e_\omega \oplus J(X)$ is the space of all scalar sequences of bounded X -variation. One may consider for any such X and any linearly ordered set I the space $V_X(I)$ of all functions $f : I \rightarrow \mathbb{R}$ of bounded X -variation. The norm on such a space is submultiplicative. The spaces $V_X[0, 1]$ were introduced in [AMP]. By very carefully combining the method of the present paper with the method from [MPZ] it is conceivable that one may obtain for every countable well ordered set I a Banach space the Calkin algebra of which

Supporting Later Result

The later paper proves the following theorem. If U has a normalized 1-unconditional basis and no c_0 asymptotic version, then there is a Banach space $\mathfrak{X}_U$ such that

$$\text{Cal}(\mathfrak{X}_U) \cong \mathbb{C}e \oplus U$$

as Banach algebras, where U is endowed with coordinate-wise multiplication. The paper also states explicitly that the earlier Problem 5 is answered by this theorem.

makes U a non-unital Banach algebra. Since every normalized unconditional basis of a Banach space U can be made 1-unconditional by passing to an equivalent norm, U can always be seen as a Banach algebra with coordinate-wise multiplication. Below, we recall the classical notion of an asymptotic version Z of a Banach space X with a basis from [40, Page 151] for the special case $Z = c_0$.

Definition 1.1. *Let X be a Banach space with a Schauder basis $(x_s)_{s=1}^\infty$. We say that X has a c_0 asymptotic version if for every $n \in \mathbb{N}$ and $\varepsilon > 0$ the following holds:*

for every $k_1 \in \mathbb{N}$ there exists $y_1 \in X$ with $\text{supp}(y_1) > k_1$,
for every $k_2 \in \mathbb{N}$ there exists $y_2 \in X$ with $\text{supp}(y_2) > k_2$,
 $\vdots$
for every $k_n \in \mathbb{N}$ there exists $y_n \in X$ with $\text{supp}(y_n) > k_n$,

such that $(y_i)_{i=1}^n$ is $(1 + \varepsilon)$ -equivalent to the unit vector basis of ℓ_∞^n .

Our paper is devoted to proving the following.

Theorem 1.2. *Let U be a Banach space with a normalized 1-unconditional basis $(u_s)_{s=1}^\infty$ not having a c_0 asymptotic version. Then, there exists a Banach space $\mathfrak{X}_U$ with a Schauder basis and a sequence of infinite-rank norm-one projections $(I_s)_{s=1}^\infty$ on $\mathfrak{X}_U$ such that the following hold.*

- (i) *The projections I_s , $s \in \mathbb{N}$, are mutually annihilating, i.e., for $s \neq t \in \mathbb{N}$, $I_s I_t = 0$.*
- (ii) *The sequence $(I_s)_{s=1}^\infty$ in $\mathcal{L}(\mathfrak{X}_U)$ and its canonical image in $\mathcal{Cal}(\mathfrak{X}_U)$ are 128-equivalent to $(u_s)_{s=1}^\infty$.*
- (iii) *$\mathcal{L}(\mathfrak{X}_U) = \mathcal{K}(\mathfrak{X}_U) \oplus [(I_s)_{s=1}^\infty] \oplus \mathbb{C}I$.*

In particular, $\mathcal{Cal}(\mathfrak{X}_U)$ is isomorphic, as a Banach algebra, to the unitization of U with coordinate-wise multiplication.

Note that because $\mathfrak{X}_U$ has a Schauder basis, it satisfies the approximation property, and thus, $\mathcal{K}(\mathfrak{X}_U)$ is the minimum non-trivial closed ideal in $\mathcal{L}(\mathfrak{X}_U)$.

The simplest examples of Banach spaces U satisfying the assumptions of Theorem 1.2 are the ℓ_p spaces, $1 \leq p < \infty$, with their unit vector bases. In particular, the unitization of the infinite-dimensional separable Hilbert space with coordinate-wise multiplication induced by an orthonormal basis is isomorphic as a Banach algebra to the Calkin algebra of some Banach space. The local property of having non-trivial cotype, implies not having a c_0 asymptotic version. The $L_p[0, 1]$ spaces, for $1 < p < \infty$, and more generally, all superreflexive rearrangement invariant spaces on $[0, 1]$ have non-trivial cotype and the Haar system as an unconditional basis (see, e.g., [35, Theorem 2.c.6]). Therefore, by Theorem 1.2, their unitizations are Calkin algebras. Due to L_1 not having an unconditional basis, Theorem 1.2 cannot be applied to it, despite it having non-trivial cotype. However, Theorem 1.2 applies to the Hardy space H_1 , which has an unconditional basis (see [39]). A non-classical Banach space with non-trivial cotype and an unconditional basis is the asymptotic- ℓ_1 Tsirelson space ([22]). There are also spaces with trivial cotype to which

the entire space $\mathfrak{X}_U$.

3. OPEN PROBLEMS AND DIRECTIONS

The most obvious question arising from our paper is whether the unitization of all Banach spaces U with an unconditional basis with coordinate-wise multiplication are Calkin algebras. The first such known example was $U = c_0$ from [44]. In [43, Problem 5], it was asked if any other such spaces existed, which is indeed true, by Theorem 1.2. Interestingly, all spaces covered by this theorem are remote from c_0 in the asymptotic sense. These two results leave a large gap between them containing such spaces with an unconditional basis as the asymptotic- c_0 Tsirelson

Proof Intuition

Problem 5 only asks for one non- c_0 unconditional-basis example. The 2024 theorem gives a large class of such examples: every eligible U contributes its coordinate-wise unitization as a Calkin algebra. The easiest witness is $U = \ell_2$, whose standard basis is unconditional and whose Banach-space geometry rules out equivalence to the c_0 basis.

Verification

Take $X = \ell_2$ with its standard orthonormal basis. This basis is normalized and 1-unconditional. The space ℓ_2 is reflexive, whereas c_0 is not, so the standard basis of ℓ_2 is not equivalent to the unit vector basis of c_0 .

Theorem 1.2 of arXiv:2401.18037 applies to $U = \ell_2$: the paper lists ℓ_p , $1 \leq p < \infty$, among the simplest examples satisfying its hypotheses and specifically mentions the infinite-dimensional separable Hilbert space case. Therefore there exists a Banach space $\mathfrak{X}_{\ell_2}$ whose Calkin algebra is isomorphic, as a Banach algebra, to the unitization of ℓ_2 with coordinate-wise multiplication. Thus $X = \ell_2$ satisfies Problem 5 of arXiv:1711.01340.

Literature and Duplicate Check

Local duplicate search over the run registry, solution packets, attempts, and active claims found no exact packet for 1711.01340 Problem 5. The existing reflexive-Calkin packet uses the same later paper for the separate published Problem 1. Web search for the exact Problem 5 phrasing and related unitization/unconditional-basis Calkin algebra terminology found the arXiv:2401.18037 answer as the decisive later source and did not reveal a newer, more specific replacement.

Limitations

This is not an original solution produced by the run. It is a literature-already-answered status packet. It also does not solve the broader problem posed in arXiv:2401.18037 asking whether every Banach space with a normalized unconditional basis has its coordinate-wise unitization realized as a Calkin algebra.